SMe Note

Acknowledge: Sample Space: Ω Random Variables: $\mathbf{Y}=(Y_1,...,Y_n)$ Sample Data: $\mathbf{y}=(y_1,...,y_n)$ $f_{\mathbf{Y}}(\mathbf{y};\theta)=\prod_i f_{\mathbf{Y}}(y_i;\theta)$,if y_i are independent Distribution Function: $F_{\mathbf{Y}}(\mathbf{y})=\mathbb{P}(Y_1\le y_1,...,Y_n\le y_n)$
Parameter Space: Θ CDF：$F_Y(y)$ PDF/PMF: $f_Y(y)$ or $p_Y(y)$

1 Basic Knowledge

Def of expectation (Moments): For any well-behaved function h on $\mathbb{R}$. $\mathbb{E}[h(Y)] := \sum_i h(y_i)\mathbb{P}(Y=y_i)$ $\mathbb{E}[h(Y)] := \int_{\mathbb{R}} h(y)f_Y(y)dy$

Def of Model: A collection of distributions indexed by a parameter $\{F_{\mathbf{Y}}(\cdot|\theta);\theta\subseteq\Theta\}$ is model.

Likelihood Function: $L(\theta;\mathbf{y})=f_{\mathbf{Y}}(\mathbf{y};\theta)$ ·**MLE**: $\hat{\theta}(\mathbf{y})=ArgMax_{\theta\in\Theta}[L(\theta;\mathbf{y})]=[U(\hat{\theta})=0]$ ps: MLE has invariant property

(Strong) Likelihood Principle: $L(\theta;x)\propto L(\theta;y)$ [Strong: $L_f(\theta;x)\propto L_g(\theta;z)$; f,g are statistical model] $\Rightarrow$ Same Inferential Conclusions.

Def of Identifiability: A model is identifiable if $f(y;\theta_1)=f(y;\theta_2),\forall y$ then $\theta_1=\theta_2$ (i.e.$map:\theta\to f(y;\theta)$ is 1-1)

・Method: **1**.By def $\Rightarrow$ ratio not depend on $y\Rightarrow\theta_1,\theta_2$ relationship. (with any additional constraint) **2**.MLE not unique $\Rightarrow$ not identifiable **3**.Give an example θ **4**.Fisher information

2 Properties of the Likelihood and the MLE

Positive [Semi-]definite matrix M: $\forall\mathbf{z}\in\mathbb{R}^n\setminus\mathbf{0},\mathbf{z}^T\mathbf{M}\mathbf{z}>0\ [\ge 0]$ $\Leftrightarrow(>0)$ [1] n positive eigenvalues. [2] full rank. [3] invertible

Score Func / Vector U	Single: $U(\theta,\mathbf{Y})=\ell'(\theta;\mathbf{Y})$	$\mathbb{E}[U]=0$	$Var[U]=\mathcal{I}(\theta)$
	Multiple: $\mathbf{U}(\theta)=[\partial_{\theta_1}\ell(\theta;\mathbf{Y}),...,\partial_{\theta_p}\ell(\theta;\mathbf{Y})]^T\in\mathbb{R}^p$	$\mathbb{E}[\mathbf{U}]=\mathbf{0}$	$Var[\mathbf{U}]=\mathcal{I}(\theta)$ $I_{j,k}=Cov(\frac{\partial\ell(\theta,\mathbf{Y})}{\partial\theta_j},\frac{\partial\ell(\theta,\mathbf{Y})}{\partial\theta_k})=-\mathbb{E}[\frac{\partial\ell(\theta,\mathbf{Y})}{\partial\theta_j},\frac{\partial\ell(\theta,\mathbf{Y})}{\partial\theta_k}]$
CLT of U	Single: $U(\theta_0;\mathbf{Y})\sim\mathcal{N}(0,\mathcal{I}(\theta_0)),_{n\to\infty}$		
	Multiple: $\mathbf{U}(\theta_0;\mathbf{Y})\sim\mathcal{N}_p(\mathbf{0},\mathcal{I}(\theta_0)),_{n\to\infty}$	Also: $\mathbf{U}(\theta_0;\mathbf{Y})^T\mathcal{I}^{-1}(\theta_0)\mathbf{U}(\theta_0;\mathbf{Y})\sim\chi^2_p$	
Fisher Information $\mathcal{I}$	Single: Observed: $J(\theta,\mathbf{y})=-U'(\theta,\mathbf{y})=-\ell''(\theta;\mathbf{y})$	Expected: $\mathcal{I}(\theta)=-\mathbb{E}[\ell''(\theta;\mathbf{Y})]$ || Unit: $i(\theta)=\mathcal{I}(\theta)/n=-\mathbb{E}[\ell''(\theta;Y_i)]$	
	Multiple: Observed: $\mathbf{J}(\theta;\mathbf{y})\in\mathbb{R}^{p\times p},J_{j,k}=-\frac{\partial^2\ell(\theta,\mathbf{y})}{\partial\theta_j\,\partial\theta_k}$	Expected: $\mathcal{I}(\theta)\in\mathbb{R}^{p\times p},\mathcal{I}_{j,k}=-\mathbb{E}[\frac{\partial^2\ell(\theta,\mathbf{Y})}{\partial\theta_j\,\partial\theta_k}]$ ps: $\mathcal{I}(\theta)$ always positive semi-definite	
$\mathcal{I}\uparrow\Rightarrow\mathbf{Y}$ 对 θ 越敏感；信息量大；参数估计更精确		A model is $identifiable$ iff $\mathcal{I}(\theta)$ is positive definite. ps:(一维) identifiable if $\mathcal{I}(\theta)\neq 0$	
CLT of MLE / CLT of I	Single: $\mathcal{I}(\theta_0)<\infty\Rightarrow\hat{\theta}\sim\mathcal{N}(\theta_0,\mathcal{I}^{-1}(\theta_0)),_{n\to\infty}$	$\widehat{Var}(\hat{\theta})\approx\mathcal{I}^{-1}(\hat{\theta})$	$ESE(\hat{\theta})\approx\sqrt{\mathcal{I}^{-1}(\hat{\theta})}$ estimated standard error
	Multiple: $\mathcal{I}(\theta_0)<\infty\Rightarrow\hat{\theta}\sim N_p(\theta_0,\mathcal{I}^{-1}(\theta_0)),_{n\to\infty}$	Also: $(\hat{\theta}-\theta_0)^T\mathcal{I}(\theta_0)(\hat{\theta}-\theta_0)\sim\chi^2_p$	

2.1 For Single Parameter

*__Random Variable / Function__: $L(\theta;\mathbf{Y})$ is random function (NOT pdf). For fixed θ, it's r.v. $\hat{\theta}(\mathbf{Y})=ArgMax_{\theta\in\Theta}[L(\theta;\mathbf{Y})]$ is r.v. ps: $\mathcal{I}^{-1},J^{-1}$ 是指其负一次方，不是反函数 True Parameter: θ_0

Name	Test Statistic	Hypothesis	CI	Other
Score Test	Given $U(\theta_0;\mathbf{Y})\sim\mathcal{N}(0,\mathcal{I}(\theta_0))$ $T_{Score}(\mathbf{Y})=\frac{U(\theta_0)}{\sqrt{\mathcal{I}(\theta_0)}}\sim\mathcal{N}(0,1)$	$H_0:\theta=\theta_0$ 不用 MLE, 泰勒展开	$\left\{\theta:\frac{U^2(\theta)}{\mathcal{I}(\theta)}<z^2_{\alpha,1}\right\}$	or $T^2_{Score}(\mathbf{Y})\sim\chi^2_1$ $T_{Score}(\mathbf{Y})\approx\frac{U(\theta_0)}{\sqrt{\mathcal{I}(\hat{\theta})}},\frac{U(\theta_0)}{\sqrt{J(\theta_0)}},\frac{U(\theta_0)}{\sqrt{J(\hat{\theta})}}$
Wald Test or Maximum Likelihood(ML) Test	$T_{Wald}(\mathbf{Y})=\sqrt{\mathcal{I}(\theta_0)}(\hat{\theta}-\theta_0)=\frac{\hat{\theta}-\theta_0}{\sqrt{\mathcal{I}^{-1}(\theta_0)}}$ $T_{Wald}(\mathbf{Y})\sim\mathcal{N}(0,1)$	$H_0:\theta=\theta_0$ 构建 CI	$\{\theta:(\hat{\theta}-\theta)^2\mathcal{I}(\theta)<z^2_{\alpha,1}\}$	or $T^2_{Wald}\sim\chi^2_1$; $_{n\to\infty}T_{Wald}\simeq T_{Score}$ $T_{Ward}(\mathbf{Y})\approx\frac{(\hat{\theta}-\theta_0)}{\sqrt{\mathcal{I}^{-1}(\hat{\theta})}},\frac{(\hat{\theta}-\theta_0)}{\sqrt{J^{-1}(\theta_0)}},\frac{(\hat{\theta}-\theta_0)}{\sqrt{J^{-1}(\hat{\theta})}}$
log-Likelihood Ratio Test	$T_{LR}(\mathbf{Y})=2(\ell(\hat{\theta})-\ell(\theta_0))\sim\chi^2_1$ as $n\to\infty$ $LR=L(\theta_0)/L(\hat{\theta})$, $T_{LR}=-2ln(LR)$	$H_0:\theta=\theta_0$ 最小 CI, Most Powerful	$\{\theta:2(\ell(\hat{\theta})-\ell(\theta))<z^2_{\alpha,1}\}$	ps: $2(\ell(\hat{\theta})-\ell(\theta_0))\simeq(\theta_0-\hat{\theta})^2J(\hat{\theta})$
$T_{Score}\simeq T_{Wald}\simeq T_{LR}$ as $n\to\infty$; $J(\theta_0)\approx\mathcal{I}(\theta_0)$ as $n\to\infty$; When $\ell(\theta)$ = 关于 θ 的二次函数 $\Rightarrow$ All same. Geometric:(T_S, 比斜率),(T_W, 比宽度),(T_{LR}，比高度) ps:$z_{\alpha,n}=\chi^2_{\alpha,n}$				

2.2 For Single Parameter (Numerical Method)

Suppose initial value solution θ_0	Newton-Raphson Method ps: also called Newton's method.	Fisher's Method of Scoring ps: also called scoring method
Formula	$\theta_{r+1}=\theta_r-\frac{U(\theta_r)}{\ell''(\theta_r)}=\theta_r+\frac{U(\theta_r)}{J(\theta_r)}$	$\theta_{r+1}=\theta_r-\frac{U(\theta_r)}{\mathbb{E}[\ell''(\theta_r)]}=\theta_r+\frac{U(\theta_r)}{\mathcal{I}(\theta_r)}$
Advantages	Converge *very quickly* near optimum	$\mathcal{I}$ is constant / easier to calculate; Converge for wider θ_0
Stoping Rule	Given tolerance level ε: [1] Stop after r if $\|U(\theta_r)\|\le\varepsilon$ (Common) or [2] Stop after $\|\theta_r-\theta_{r-1}\|\le\varepsilon$	

2.3 For Multiple Parameters

* 此节中，θ 代表向量,$\theta=(\theta_1,...,\theta_p)$ $\mathcal{I}^{-1}$ 指其逆矩阵 True Parameter: θ_0

Name	Test Statistic ($T^2_{Score},T^2_{Wald},T_{LR}$)	Hypothesis	(Confidence Region) CR
Score Test	$\mathbf{U}(\theta_0)^T\mathcal{I}^{-1}(\theta_0)\mathbf{U}(\theta_0)>z_{\alpha,p}$	$\theta=\theta_0$	$\{\theta:\mathbf{U}(\hat{\theta})^T\mathcal{I}^{-1}(\theta)\mathbf{U}(\hat{\theta})<z_{\alpha,p}\}$
Wald Test	$(\hat{\theta}-\theta_0)^T\mathcal{I}(\theta_0)(\hat{\theta}-\theta_0)>z_{\alpha,p}$	$\theta=\theta_0$	$\{\theta:(\hat{\theta}-\theta)^T\mathcal{I}(\theta)(\hat{\theta}-\theta)<z_{\alpha,p}\}$
log-Likelihood Ratio Test	$2(\ell(\hat{\theta})-\ell(\theta_0))>z_{\alpha,p}$	$\theta=\theta_0$	$\{\theta:2(\ell(\hat{\theta})-\ell(\theta))<z_{\alpha,p}\}$
$\mathcal{I}(\theta_0)\approx\mathcal{I}(\hat{\theta}),\mathbf{J}(\theta_0),\mathbf{J}(\hat{\theta})$ $ps:z_{\alpha,p}=\chi^2_{\alpha,p}$ T_{LR}:simple hypothesis testing.($dim(\mathcal{A})=0$) θ 都是向量			

Marginal Normal distribution $\hat{\theta}$:

$\hat{\theta}_j\sim\mathcal{N}(\theta_j,[\mathcal{I}^{-1}(\underline{\theta_0})]_{jj})$ θ_j: true para

·$ESE(\hat{\theta}_j)=\sqrt{[\mathcal{I}^{-1}(\hat{\theta})]_{jj}}$

· CI(θ_j): $\hat{\theta}_j\pm z_\alpha ESE(\hat{\theta}_j)$

Composite Hypothesis Test: $H_0:\underline{\theta}\in\mathcal{A},H_1:\underline{\theta}\notin\mathcal{A}$ $\mathcal{A}$: reduced model $LR=\frac{max_{\underline{\theta}\in\mathcal{A}}L(\underline{\theta})}{max_{\underline{\theta}\in\Theta}L(\underline{\theta})}=\frac{L(\hat{\underline{\theta}}_{\mathcal{A}})}{L(\hat{\underline{\theta}}_{\Theta})}$ ps: 多维未知参数,H_0 只想其中几个时用 \|\| $dim(\mathcal{A})$ 常数时为 0

· **Generalised Likelihood Ratio Test**: $T_{GLR}=-2ln(LR)=2\left(\ell(\hat{\underline{\theta}})-\ell(\hat{\underline{\theta}}_{\mathcal{A}})\right)\sim\chi^2_{p-r}$ p : # of parameters. $r=dim(\mathcal{A})$

2.4 For Multiple Parameter (Numerical Method)

Method (start at θ_0) * 此节中，θ 代表向量	Newton-Raphson Method	Fisher's Method of Scoring
Formula	$\theta_{r+1}=\theta_r-\mathbf{H}^{-1}(\theta_r)\mathbf{U}(\theta_r)=\theta_r+\mathbf{J}^{-1}(\theta_r)\mathbf{U}(\theta_r)$	$\theta_{r+1}=\theta_r-(\mathbb{E}[\mathbf{H}(\theta_r)])^{-1}\mathbf{U}(\theta_r)=\theta_r+\mathcal{I}^{-1}(\theta_r)\mathbf{U}(\theta_r)$
ps: Jacobian / Hessian $\mathbf{H}=-\mathbf{J}$ || If $\mathcal{I}=diag(a,b,...)$,$\Rightarrow\mathcal{I}^{-1}=diag(a^{-1},b^{-1},...)$		

3 Bayesian Inference

3.1 Bayesian Approach & Definitions

Basic Idea: Consider $\underline{\theta}$ as a r.v.. Bayesian approach ⇒ Probability are used to represent degree of believe about the unknown $\underline{\theta}$

Prior Distribution $p(\underline{\theta})$: Prior distribution $p(\underline{\theta})$ contains all available prior information about $\underline{\theta}$ *before observing any data*.

· **Hyper-Parameters**: 由于 $p(\underline{\theta})$ 是已知, 服从某种分布 (with parameter γ). We denoted $p(\underline{\theta})$ as $p(\underline{\theta};\gamma)$ where γ is *Hyper-parameters*. 定义: $\gamma_{prior} \to \gamma_{post}$

Likelihood function $p(\mathbf{y}|\underline{\theta}) = f(\mathbf{y}|\underline{\theta})$: Any observed data is considered as a realisation of r.v. Y with $f(\mathbf{y}|\underline{\theta})$

Posterior Distribution $p(\underline{\theta}|\mathbf{y})$: Posterior distribution $p(\underline{\theta}|\mathbf{y})$, contains all available knowledge.

· **Bayes' Rule / Theorem**: $p(\underline{\theta}|\mathbf{y}) = \frac{p(\mathbf{y}|\underline{\theta})p(\underline{\theta})}{p(\mathbf{y})} = \frac{f(\mathbf{y}|\underline{\theta})p(\underline{\theta})}{m(\mathbf{y})}$ $p(\mathbf{y}) = m(\mathbf{y}) = \int_{\underline{\theta}\in\Theta} f(\mathbf{y}|\underline{\theta})p(\underline{\theta})d\underline{\theta}$ (Normalizing constant)

Name	MAP Maximum A Posteriori	MMSE Posterior Mean / Bayesian Minimum Mean Squared Error Estimator	Posterior Median $\hat{\theta}_{median}$
Single Para	$\hat{\theta}_{MAP} = \arg\max_\theta p(\theta|\mathbf{y})$	$\hat{\theta}_{MMSE} = \mathbb{E}[\theta|\mathbf{y}] = \int_{\theta\in\Theta} \theta p(\theta|\mathbf{y})d\theta$	$\frac{1}{2} = \int_{-\infty}^{\hat{\theta}_{median}} p(\theta|\mathbf{y})d\theta = \int_{\hat{\theta}_{median}}^{\infty} p(\theta|\mathbf{y})d\theta$

Name	Posterior Var	$C_\alpha = [a,b]$ $(1-\alpha)100\%$ Bayesian Credible Intervals (or Simply Credible Intervals)
Single Para	$Var[\theta|\mathbf{y}] = \int_{\theta\in\Theta}(\theta - \mathbb{E}[\theta|\mathbf{y}])^2 p(\theta|\mathbf{y})d\theta$	$\mathbb{P}(a \le \theta \le b|\mathbf{y}) = \int_a^b p(\theta|\mathbf{y})d\theta = 1-\alpha$ ps: $\mathbb{P}(\theta < a|\mathbf{y}) = \mathbb{P}(\theta > b|\mathbf{y}) = \alpha/2$

Bayesian Decision Theory: 目的: [1]Estimating parameter θ [2]Test Hypothesis about θ [3]Find CI. ps: 后两种有很多种方法来判断

1. Choose a *Loss Function* $Q(\hat{\theta},\theta)$ (损失函数, 衡量估计值与真实值的差距; 贝叶斯决策的目的是最小化损失函数的期望)
2. Minimises Posterior Expacted Loss Function: $\hat{\theta}_Q(\mathbf{y}) = \arg\min_{\hat{\theta}} \mathbb{E}[Q(\hat{\theta},\theta)|\mathbf{y}] = \int_{\theta\in\Theta} Q(\hat{\theta},\theta)p(\theta|\mathbf{y})d\theta$

3.2 Exact Bayesian Inference with Conjugate priors

Conjugate Prior: For $\underline{\theta}, p(\underline{\theta})$ is *Conjugate Prior* for $p(\mathbf{y}|\underline{\theta}) = f(\mathbf{y}|\underline{\theta})$ iff 对 $p(\underline{\theta})$ 用 Bayes' Rule 得到的 $p(\mathbf{y}|\underline{\theta})$ 都在同一类型的分布

Conjugate Family: The collection of all *Conjugate Prior*.

Choice of Prior: Observed data number n: [1]n small ⇒ Posterior shows a great dependency of the choice of prior.

[2]n middle ⇒ posterior less affected by choice of prior. [3]n large ⇒ choice of prior has limited effect on posterior.(Likelihood has a strong impact)

Comparing the frequentist and Bayesian approaches: [1] $\hat{\theta}_{MMSE} \approx \hat{\theta}_{MAP}$ for most cases, especially n large. [2]n large: $\hat{\theta}_{MAP} \approx \hat{\theta}_{MLE}$, $CI \approx C_\alpha$

3.3 Approximate Bayesian Inference

Intractable Posterior: 算不出具体分布或不 conjugate 的 model. For some models, the posterior distribution is intractable.

4 Distribution Table

Distribution	PDF	Mean	Var	Other
Bernoulli $X \sim Bern(p)$	$p^x(1-p)^{1-x}, x \in \{0,1\}$	p	$p(1-p)$	If X_i are i.i.d. $\sum X_i \sim Bin(n,p)$
Binomial $X \sim Bin(n,p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	np	$np(1-p)$	$X,Y \sim Bin(n,p), Bin(m,p)$ $X+Y \sim Bin(n+m,p)$
For $X \sim Bin(n,p), n\to\infty, p\to 0, np\to\lambda$, then $X \sim Poission(\lambda)$		For $X \sim Bin(n,p), n\to\infty$ and pdf symmetric, ⇒ $X \sim N(np, np(1-p))$		
Geometric $X \sim Geom(p)$	$(1-p)^{x-1}p$	$1/p$	$(1-p)/p^2$	$\mathbb{E}[X^2] = 2/p^2$
Negative Binomial $X \sim NB(r,p)$	$\binom{r+x-1}{x}(1-p)^r p^x$, $r \ge 1$	$\frac{rp}{1-p}$	$\frac{rp}{(1-p)^2}$	或者 $p = 1-q$, 注意!
Poission $X \sim Po(\lambda)$	$e^{-\lambda}\cdot\frac{\lambda^x}{x!}$	λ	λ	$\mathbb{E}[X^2] = \lambda^2 + \lambda$
For $X,Y \sim Po(\lambda_1), Po(\lambda_2)$, then $X+Y \sim Po(\lambda_1+\lambda_1)$				
Multinomial $x_1+\cdots+x_k = n$ $(X_1,...,X_k) \sim MN(n,p_1,...,p_k)$	$\frac{n!}{x_1!\cdots x_k!}p_1^{x_1}\cdots p_k^{x_k}$, $0<p_i<1$	np_i	$np_i(1-p_i)$	
Uniform $X \sim \mathcal{U}([a,b])$	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\mathbb{E}[X^2] = \frac{b^2+ab+a^2}{3}$
Normal $X \sim \mathcal{N}(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}}\cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	$\mathbf{Y} \sim N_k(\mu,\Sigma)$, μ vector $f_{\mathbf{Y}}(y_1,...,y_k) = \frac{e^{-\frac{1}{2}(y-\mu)^T\Sigma^{-1}(y-\mu)}}{\sqrt{(2\pi)^k det(\Sigma)}}$
For $X,Y \sim N(\mu_1,\sigma_1^2), Po(\mu_2,\sigma_2^2)$, then $aX \pm Y \pm c \sim Po(a\mu_1 \pm b\mu_2 \pm c, a^2\sigma_1^2 + b^2\sigma_2^2)$				
Exponential $X \sim Exp(\lambda)$	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\mathbb{E}[X^n] = \frac{n!}{\lambda^n}$
Beta $X \sim Beta(\alpha,\beta), \alpha,\beta > 0$	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}, x \in [0,1]$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$
When $Beta(\alpha,\beta)$ has maximum probability, $x = (\alpha-1)/(\alpha+\beta-2)$				
Gamma $X \sim \Gamma(\alpha,\beta), \alpha,\beta > 0$	$\frac{\beta^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\beta x}, x \ge 0$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$	$\Gamma(\alpha) = (\alpha-1)!$ $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1}e^{-x}dx$
Student's t $X \sim t_v, v > 0$	$\frac{1}{\sqrt{v}B(\frac{v}{2},\frac{1}{2})}(1+\frac{x^2}{v})^{-\frac{v+1}{2}}$	0	$\frac{v}{v-2}$	If $Z \sim N(0,1), Y \sim \chi_n^2$ $\frac{Z}{\sqrt{Y/n}} \sim t_n$
PDF Symmetric at zero and has more mass in the tails (compare to the standard normal distribution) When $n\to\infty$ it will converges to $N(0,1)$. **Cauchy Distribution**: $X \sim t_1$ and its mean and var are undefined.				

Chi-Squared $X \sim \chi^2_v, v > 0$	$\frac{1}{2^{\frac{v}{2}}\Gamma(\frac{v}{2})} x^{\frac{v}{2}-1} e^{-\frac{x}{2}}, x \geq 0$	v	$2v$	-
If $X_i \sim N(0,1),\ then\ \sum X_i^2 \sim \chi^2_n$	If $X \sim \chi^2_n, Y \sim \chi^2_m,\ then\ X+Y \sim \chi^2_{n+m}$			
F $X \sim F(\lambda, v), v > 0$ ps: $f_{a,b;\alpha} = 1/f_{a,b;1-\alpha}$	$\frac{\lambda^{\frac{\lambda}{2}} v^{\frac{v}{2}}}{B(\frac{\lambda}{2},\frac{v}{2})} x^{\frac{\lambda}{2}-1} (\lambda x + v)^{-\frac{\lambda+v}{2}}, x \geq 0$	$\frac{v}{v-2}$	$\frac{2v^2(\lambda+v-2)}{\lambda(v-2)^2(v-4)}, v > 4$	$Y_1, Y_2 \sim \chi^2_{n_1}, \chi^2_{n_2}, X = \frac{Y_1/n_1}{Y_2/n_2}$ $X \sim F_{n_1,n_2}, X^{-1} \sim F_{n_2,n_1}$

5 Review

Law of Total Probability: $\mathbb{P}(A) = \sum_n \mathbb{P}(A|B_n)\mathbb{P}(B_n)$, $\cup_n B_n = \Omega$, B_n disjoint

Type	Type I error	Type II error
Def	$\mathbb{P}(Rej\ H_0 \mid H_0\ is\ T)$	$\mathbb{P}(Fail\ Rej\ H_0 \mid H_0\ is\ F)$

Significance Level: $\alpha = \mathbb{P}(Rej\ H_0 \mid H_0\ is\ T)$

-	H_0 is T	H_0 is F
Fail to Rej H_0	Right	Type II (β) 与 size 有关
Rej H_0	Type I (α) 与 size 无关	Right (Power)

P-value: 1.One-side: $p = \mathbb{P}(T \geq t; T \sim H_0)$ 2.Two-sides: $p = 2\min\{\mathbb{P}(T \geq t; T \sim H_0), \mathbb{P}(T \leq t; T \sim H_0)\}$

Power of Test: $\mathbb{P}(\text{Rej } H_0 | H_0 \text{ is F}) = 1 - \beta = \mathbb{P}(Z \in C | \mu = \mu^*)$ · 1.Sample size ↑, Power ↑; 2.σ^2 ↓, Power ↑; 3.Effect size $|\theta_0 - \theta^*|$ ↑, Power ↑; 4.α↑, Power ↑.

Def of Confidence Interval (CI): $100(1-\alpha)\%$ Confidence Interval for θ is $[l(X), u(X)]$, $\mathbb{P}(l(X) < \theta < u(X)) = 1 - \alpha$.

(Convergence of r.v.) Name	Definition	written as	Example	
Convergence in Probability	If $\lim F_{X_n}(x) = F_X(x)$ for all $x \in \mathbb{R}$.	$X_n \xrightarrow{d} X$ or $X_n \xrightarrow{d} F_X$	CLT 最弱:	见下面
Convergence in Distribution	If $\forall \epsilon > 0, \lim \mathbb{P}(\lvert X_n - X \rvert > \epsilon) = 0$.	$X_n \xrightarrow{p} X$	WLLN 中等:	If $X_i \overset{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$. $\Rightarrow\ \bar{X}_n \xrightarrow{p} \mu$
Almost Sure Convergence	If $\mathbb{P}(\lim X_n = X) = 1$	$X_n \xrightarrow{a.s.} X$	SLLN 最强:	If $X_i \overset{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$, $Var[X_i] < \infty. \Rightarrow\ \bar{X}_n \xrightarrow{a.s.} \mu$

Central Limit Theorem (CLT): If $X_i \overset{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$ and $Var[X_i] < \infty$. $\Rightarrow\ \frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \overset{d}{\sim} N(0,1)$ 样本均值的分布会趋近于正态分布

6 Writting Format

Distribution: 产生新 distribution 的一种方法: 通过条件概率来. $\mathbb{P}(A|B) = \mathbb{P}[(A \cap B)/B]$

p-value: p-value measures the plausibility that the data were generated under H_0. **Low p-value**: low p-value suggests H_0 is incompatible with the data and hence likely to be false.

CI: The set of values (e.g.θ) are not rejected at significance level α under H_0. If data are generated under H_0, there is a probability $(1-\alpha)$ for the true θ_0 to be within the $(1-\alpha)100\%$ CI.

Linear Regression Assumptions: Assumptions are that the obervations: 1. are from a normal distribution; 2. are independent; 3. have a common variance; 4. Linearity of the expectation variable.

Hypothesis Testing: **Steps**: ^{1} State H_0, H_1; ^{2} Select Test Statistic T; ^{3} distribution of T under H_0; ^{4} observed test statistic; ^{5} critical value / region, p-value, or CI; ^{6} conclusion. **Reject H_0**: We would reject H_0 in favour of the H_1. (So that ...) **Fail to reject H_0**: We fail to reject H_0. Hence, there is strong evidence against the null hypothesis.

MLE: **Using (Compare) Newton's Method = (≠) Scoring Method**: Because the second derivative of likelihood function doesn't (does) include the r.v.

MLE Inference Parameter|$\mathbb{P}(\theta < ?) = \alpha$ is false: We cannot make such probability statement. $\{\theta\}$ is a fixed value (but unknown).

Geoemtric Comparison of T_{Score},T_{Wald},T_{LR}: **1.**T_{Score} (Slope Comparison): Represents the slope of the likelihood function at θ_0; A steeper slope indicates that the likelihood function is changing more rapidly at $theta_0$, which implies a greater difference between $\hat{\theta}$ and θ_0. **2.**T_{Wald} (Width Comparison): Compares the difference between $\hat{\theta}$ and θ_0. **3.**T_{LR} (Height Comparison): Compares the maximum likelihood value $L(\hat{\theta})$ and the likelihood value at θ_0, $L(\theta_0)$. A larger height difference indicates a greater discrepancy between $\hat{\theta}$ and θ_0 T_{Score},T_{Wald},T_{LR} **Other** 部分公式可以近似的理由: This is due to $\hat{\theta}$ being a consistent estiamtor of θ_0, and $J(\theta_0)$ being a consistent estimator of $\mathcal{I}(\theta_0)$